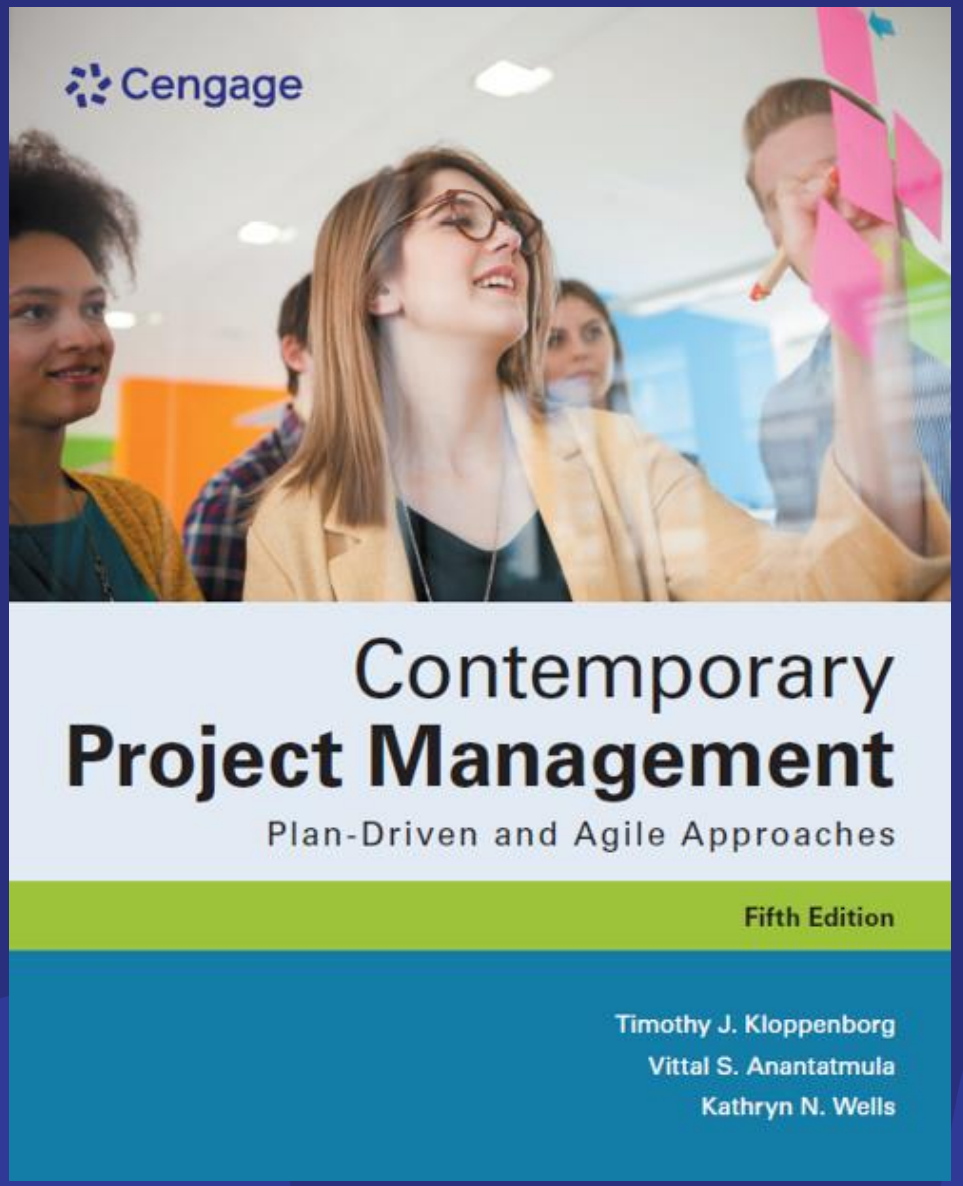


Chapter 12

Project Quality Planning and Project Kickoff

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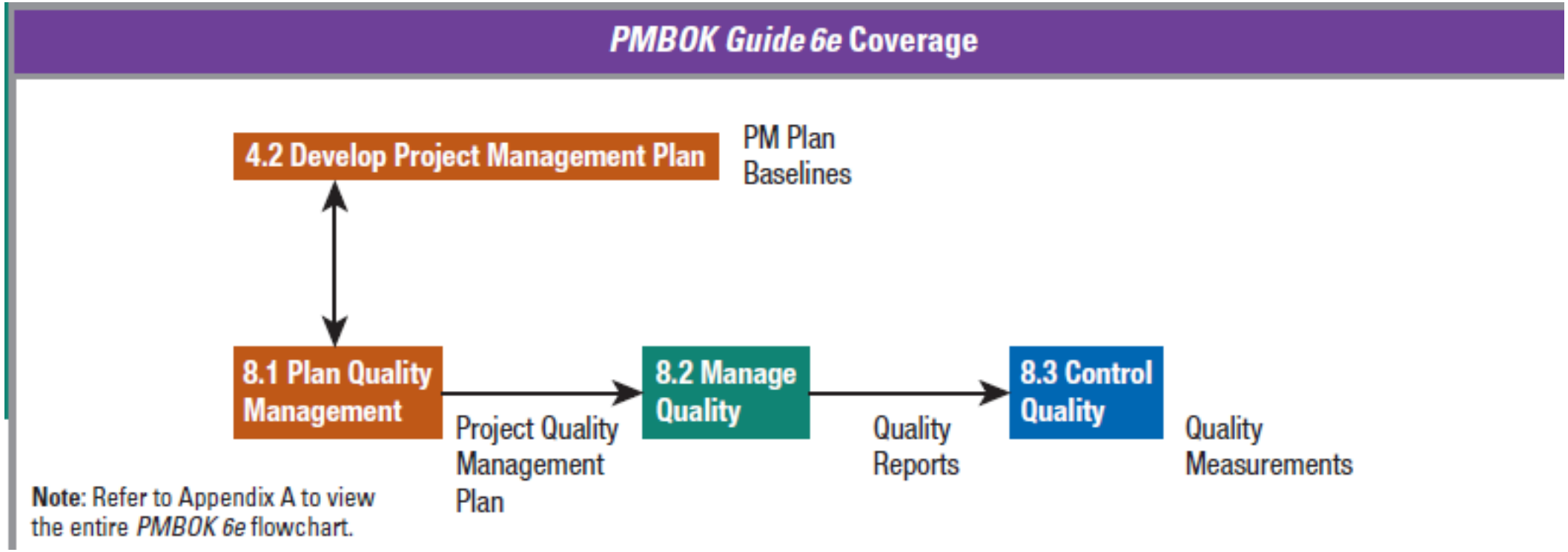
Icebreaker

What are two things you learned from Project Risk Planning?



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Project Quality Planning and Project Kickoff (1 of 2)



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Project Quality Planning and Project Kickoff (2 of 2)

Overview of the Book
12.1 Development of Contemporary Quality Concepts
12.2 Core Project Quality Concepts
12.3 Plan Quality Management
12.4 Manage Quality
12.5 Control Quality
12.6 Cost of Quality
12.7 Develop Project Management Plan
12.8 Kickoff Project
12.9 Baseline and Communication Project Management Plan
12.10 *PMBOK Guide 7e*
12.11 Agile Projects
12.12 Using MS Project for Project Baselines

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Core Objectives

- **12-1** Define each core concept of quality in projects and explain its significance in planning and managing projects.
- **12-2** Explain what may be included in a project quality management plan.
- **12-3** Describe the major contributions to contemporary project quality of the quality gurus and TQM, ISO, and Six Sigma.
- **12-4** Baseline your complete project plan in Microsoft Project.
- **12-5** Compile a complete project management plan, including all parts covered in the last several chapters.
- **12-6** Kick off a project with effective pre-meeting preparation, a kickoff meeting, and documentation.

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Agile Objective

- **12-7** Describe how Agile methods help to improve the quality of project products, processes, communications, and people.

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Project Quality Management

Plan Quality Management is "the process of identifying quality requirements and/or standards for the project and its deliverables and documenting how the project will demonstrate compliance."

- PMBOK Guide 6th Edition

- All the necessary work to ensure the project deliverables satisfy their intended purpose.
- The first step in project quality management is **plan quality management**.
- May be performed simultaneously with other aspects of project planning.

In Agile, quality is planned at a high level for the entire project at the outset and at a detailed level just before the start of each iteration.

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12.1 Development of Contemporary Quality Concepts

12.1a Quality Gurus
12.1b Total Quality Management/Malcolm Baldrige
12.1c ISO 9001:2015
12.1d Lean Six Sigma

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Quality Guru—*W. Edwards Deming*

W. Edwards Deming summarized his ideas as follows:

- Understanding variation is essential to improving quality.
- It is important to understand how companies operate as systems.
- Managers need insight in order to accurately predict the future.
- Leaders need to understand individual motivations.
- Developed a system of profound knowledge based on these ideas.

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Exhibit 12.1

Deming's Profound Knowledge System

Systems	Interactions occur among parts of a system, and parts cannot be managed in isolation.
Variation	Managers need to understand common and special causes of variation and then work to reduce both.
Knowledge	Managers need to learn from the past and understand cause-and-effect relationships to predict future behavior.
Psychology	Leaders need to understand what motivates each individual and how different people and groups interact.

Source: Adapted from James R. Evans and William M. Lindsay, *The Management and Control of Quality*, 8th ed. (Mason, OH: Cengage Learning South-Western, 2011): 94–99.

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Quality Guru—*Joseph Juran*

- Joseph Juran, who was a contemporary of Deming, also wrote and lectured prolifically for decades.
- Juran is perhaps best known for his Quality Trilogy of quality planning, quality control, and quality improvement.

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Exhibit 12.2

Juran's Quality Trilogy

Quality Planning	Identify all customers and their needs, develop requirements based upon those needs, and develop the methods to satisfy those requirements.
Quality Control	Determine what to control, establish measurement systems, establish standards, compare performance to standards, and act on differences.
Quality Improvement	Select and support improvement projects, prove causes, select and implement solutions, and maintain control of improved processes.

Source: Adapted from James R. Evans and William M. Lindsay, *The Management and Control of Quality*, 8th ed. (Mason, OH: Cengage Learning South-Western, 2011): 104–106.

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Exhibit 12.3

Plan-Driven Project Roles

Thought Leader	Additional Key Project Quality Contributions
Clifton	High-quality organizations encourage individuals to develop their strengths.
Crosby	Quality is meeting requirements, not exceeding them. The burden of quality falls on those who do the work. Quality costs least when work is done correctly the first time. Quality improves more by preventing defects rather than fixing them.
Harrington	Business processes can be improved using a systematic method.
Ishikawa	Quality outputs start with understanding customers and their desires. Work to identify and remove root causes, not just symptoms. All workers at all levels must engage to improve quality. Most quality problems can be solved by using simple tools.
Senge	Team learning is necessary to improve quality.
Shiba	Societal networking accelerates quality improvement. When continual improvement is not enough, a breakthrough is needed.
Taguchi	Reducing variation saves money. Project deliverables will be better with a focus on improving methods.
Zeithaml	Services pose different challenges from manufacturing when improving quality.

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12.1b Total Quality Management/Malcolm Baldrige

- In the 1980s, it was evident that inspection was not an adequate means of discovering quality problems.
- Core concepts of TQM (**Total Quality Management**) evolved into criteria for the Malcolm Baldrige Performance Excellence Award.
- Include concepts such as leadership, customer focus, learning, partnership, innovation, and managing by facts.

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Exhibit 12.4

Malcom Baldrige Performance Excellence Award Core Values

Core Value	Brief Interpretation of Core Value
Systems perspective	Manage organization as unified whole
Visionary leadership	Demonstrate clear values and set high expectations
Customer focused excellence	Consider all characteristics that contribute to customer success
Valuing people	Value workforce and all stakeholders
Agility and resilience	Able to anticipate, prepare for, and recover from disruptions
Organizational learning	Continuously improve, and significantly change, leading to new goals
Focus on success and innovation	Drive change in your products, services, and operations to create value
Management by fact	Measure and analyze performance to support decision making
Societal contribution	Be role models for well-being of your communities
Ethics and transparency	Stress ethical behavior and transparency in all interactions Delivering value
Delivering value and results	Balance finance and process results, customer and worker satisfaction

Source: Adapted from: https://www.nist.gov/baldrige/core-values-and-concepts. Baldrige Performance Excellence Program Core Values and Concepts. Accessed October 25, 2021.

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12.1c ISO 9001:2015

- A quality framework developed in Europe.
- Developed by the International Organization for Standardization.
- **ISO 9001** is the quality management standard.
- **2015** represents the latest revision of the standard.
- Originally focused on technical standards in documenting work processes.
- Evolved into a comprehensive set of principles.

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Exhibit 12.5

ISO Quality Management Principles

Principle	Brief Description of Principle
Customer focus	Meet customer requirements and exceed customer expectations
Leadership	Establish unity of purpose and create conditions for people to succeed
Engagement of people	Competent, empowered, and engaged people create and deliver value
Process approach	Understand and manage processes to produce desired results
Improvement	Successful organizations consistently focus on improvement
Evidence-based decision making	Decisions based on analysis and evaluation produce desired results
Relationship management	Manage relationships with all interested parties including suppliers

Adapted from: https://www.iso.org/publication/PUB100080.html. ISO Quality Management Principles. Accessed October 25, 2021.

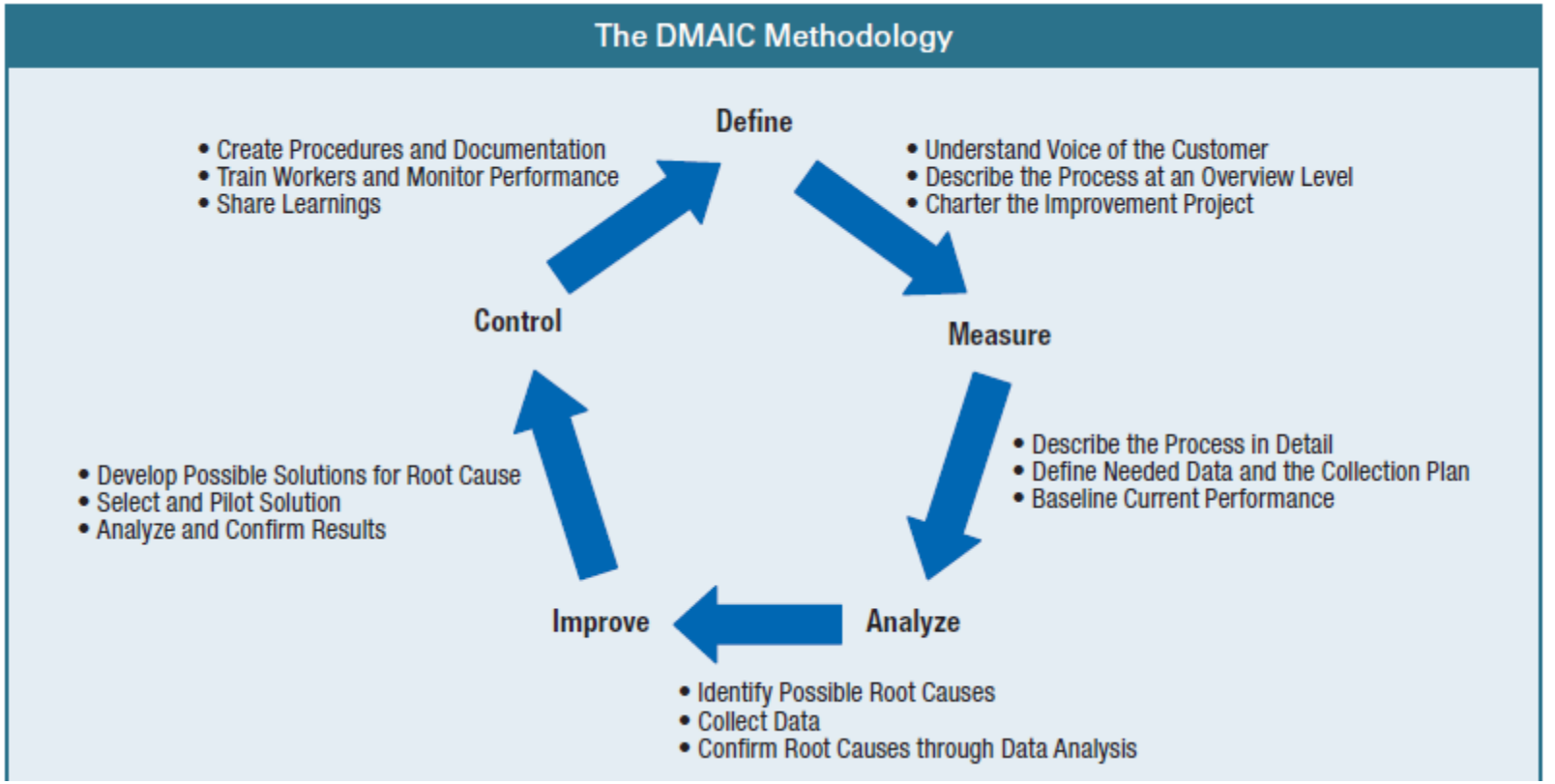
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12.1d Lean Six Sigma

- **Sigma** stands for **standard deviation**—*a statistical term for the amount of variation in data.*
- Quality problems are measured in parts per million opportunities.
- Rigor of statistics in Six Sigma is not always applicable.
- Six Sigma uses a disciplined process DMAIC to plan and manage improvement projects.

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Exhibit 12.6



Source: Timothy J. Kloppenborg and Laurence J. Laning, *Strategic Leadership of Portfolio and Project Management* (New York: Business Expert Press, 2012), 122.

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12.2 Core Project Quality Concepts (1 of 2)

12.2a Stakeholder Satisfaction
12.2b Process Management
12.2c Fact-Based Management
12.2d Fact-Based Project Management Example
12.2e Empowered Performance
12.2f Summary of Core Concepts

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12.2 Core Project Quality Concepts (2 of 2)

When defining quality, you must consider the following perspectives:

- **Product**—*presence of desired attributes.*
- **Value**—*ratio of benefits to price.*
- **Manufacturing**—*consistency in goods and services.*
- **Customers**—*ability to satisfy given needs and expectations.*

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12.2a Stakeholder Satisfaction

- Identify all stakeholders:
  - *External stakeholders—customers, suppliers, the public.*
  - *Internal stakeholders—shareholders and workers at all levels.*
- Determine relevant quality standards.
- Understand ultimate quality goals with respect to stakeholders:
  - *Project manager may facilitate a process in which important stakeholders use the quality standards to judge the project process and deliverables.*

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Developing Quality Standards Based upon Stakeholder Requirements

1. Identify all stakeholders.
2. Prioritize among the stakeholders.
3. Understand the prioritized stakeholders' requirements.
4. Develop standards to ensure the requirements are met.
5. Make tradeoff decisions.

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Stakeholder Satisfaction Sayings

"Measure twice, cut once."

"Meet requirements, but exceed expectations."

"A smart project manager develops capable customers."

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12.2b Process Management

Process is a sequence of connected activities whose goal is to produce a good, service, or specific result.

To effectively manage project processes, project managers need to understand, control, and improve them:

- *Process Understanding with a SIPOC Model.*
- *Process Control.*
- *Process Improvement with a PDCA Model.*

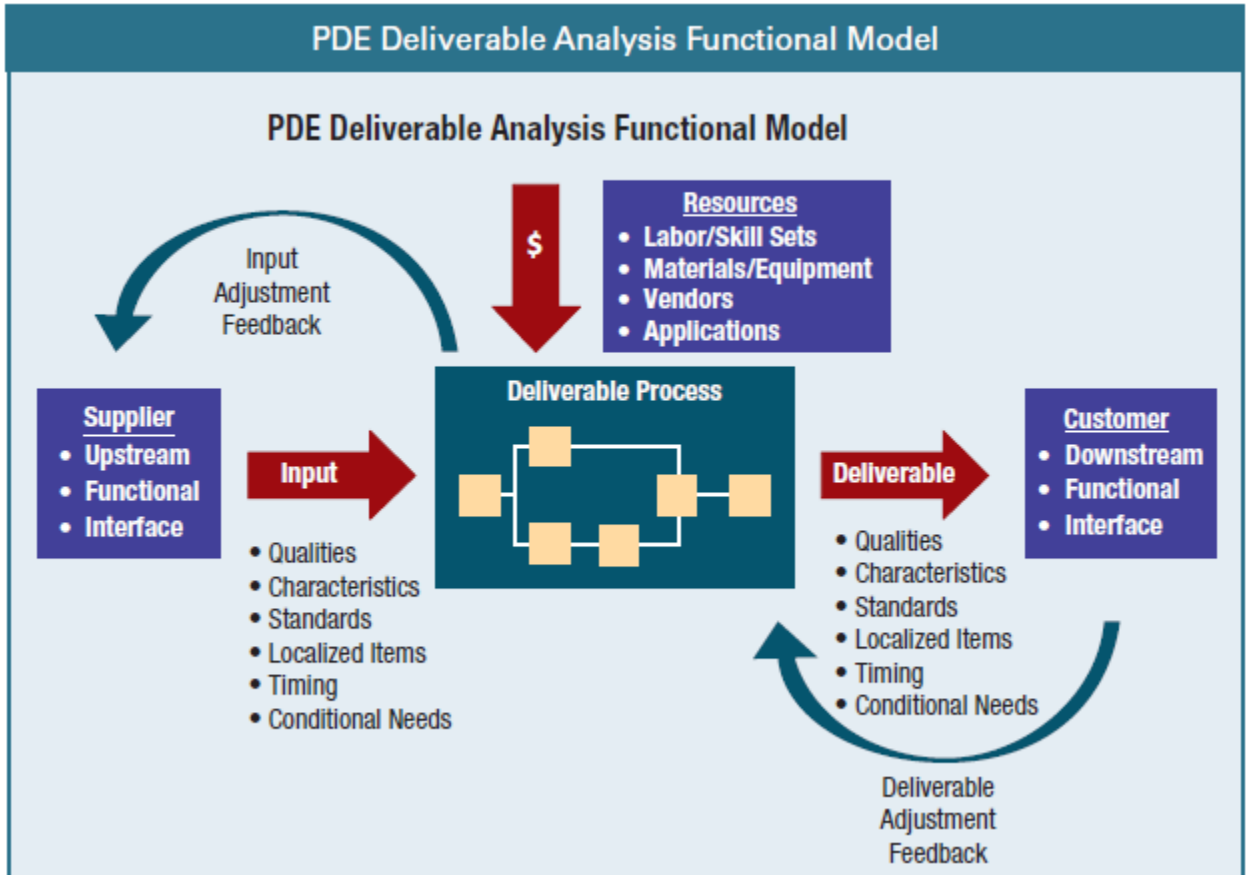
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Process Understanding with a SIPOC Model

- All work flows from suppliers, through the project, to customers.
- Use a **Supplier-Input-Process-Output-Customer** (SIPOC) model to envision this flow.
- Think backward from the project's customer.
- Determine if the process is capable of creating project deliverables.
- Design quality into the processes.

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Exhibit 12.7



Source: Paul Kling, PMP.

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Process Control

In order to have confidence that outputs are predictable:

- Establish standards.
- Measure actual performance.
- Calculate difference between standards and performance *(variance)*.
- Strive to reduce unwanted variance.

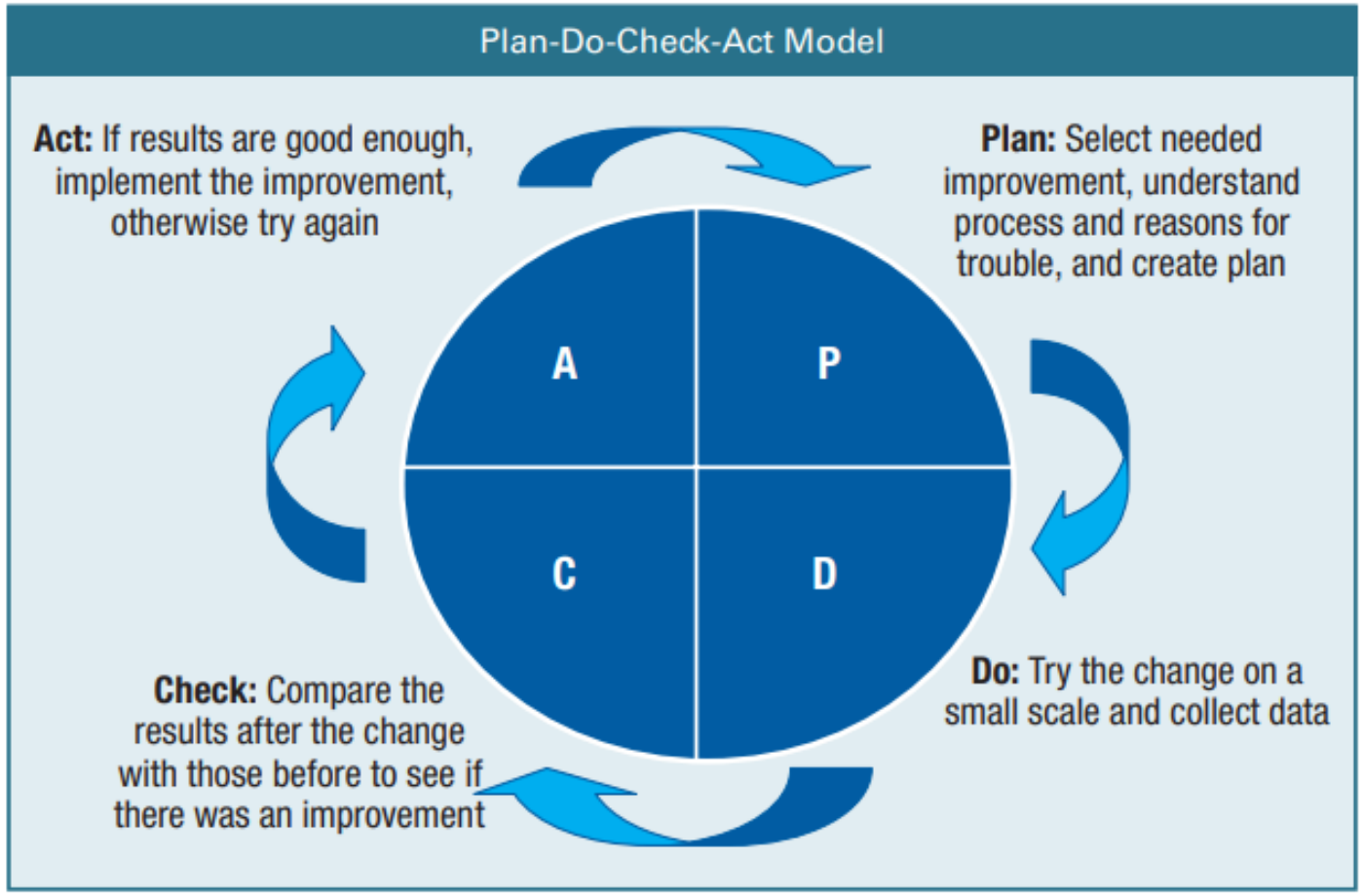
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Process Improvement with a PDCA Model

- Processes can be improved in a continuous or breakthrough fashion:
  - *Slow and steady improvement is a good foundation.*
  - *Substantial improvement requires a breakthrough.*
- Models based on scientific method exist to guide the improvement process.
- Commonly used models include the DMAIC used by Six Sigma or the Plan-Do-Check-Act (PDCA) model.

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12.2c Fact-Based Management

- Opinions get in the way.
- Project decisions may need to be made quickly.
- Taking just enough time to get the facts often yields better results.

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Four Aspects of Fact-Based Management

1. Understanding variation.
2. Deciding what to measure.
3. Working correctly with data.
4. Using the resulting information appropriately.

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12.2c Understanding Variation

Common Cause

Within the range of what can be expected.

vs.

Special Cause

Something unusual is happening

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Determining What to Measure

A Quality Metric is what to measure *(and how)* to ensure customer satisfaction.

- Avoid not measuring anything.
- Avoid measuring many things just to be sure.
- Refer to lessons learned from previous projects.
- Acceptance criteria from milestone schedule is a good starting point.
- Agree on what measures, when, and under what circumstances.

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Working Correctly with Data

- Devise simple methods of collecting and analyzing data.
- The person closest to the situation should collect the data.
- Ensure data are complete, accurate, timely.
- Turn raw data into information useful to decision makers.

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Using the Resulting Information Appropriately

- Encourage truth and transparency in communications.
- Use information to challenge opinions and decisions.
- Project communication plan outlines how to share information.

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12.2d Fact-Based Project Management Example

A rapidly growing, fast, casual restaurant chain was experiencing growing pains that were manifest in delays for restaurant openings.

- **SWOT** analysis revealed several broad issues that were threats to opening the restaurants on time, but the issues needed to be quantified for improvement.
- The company had been keeping records on the time required to open a restaurant after breaking ground.

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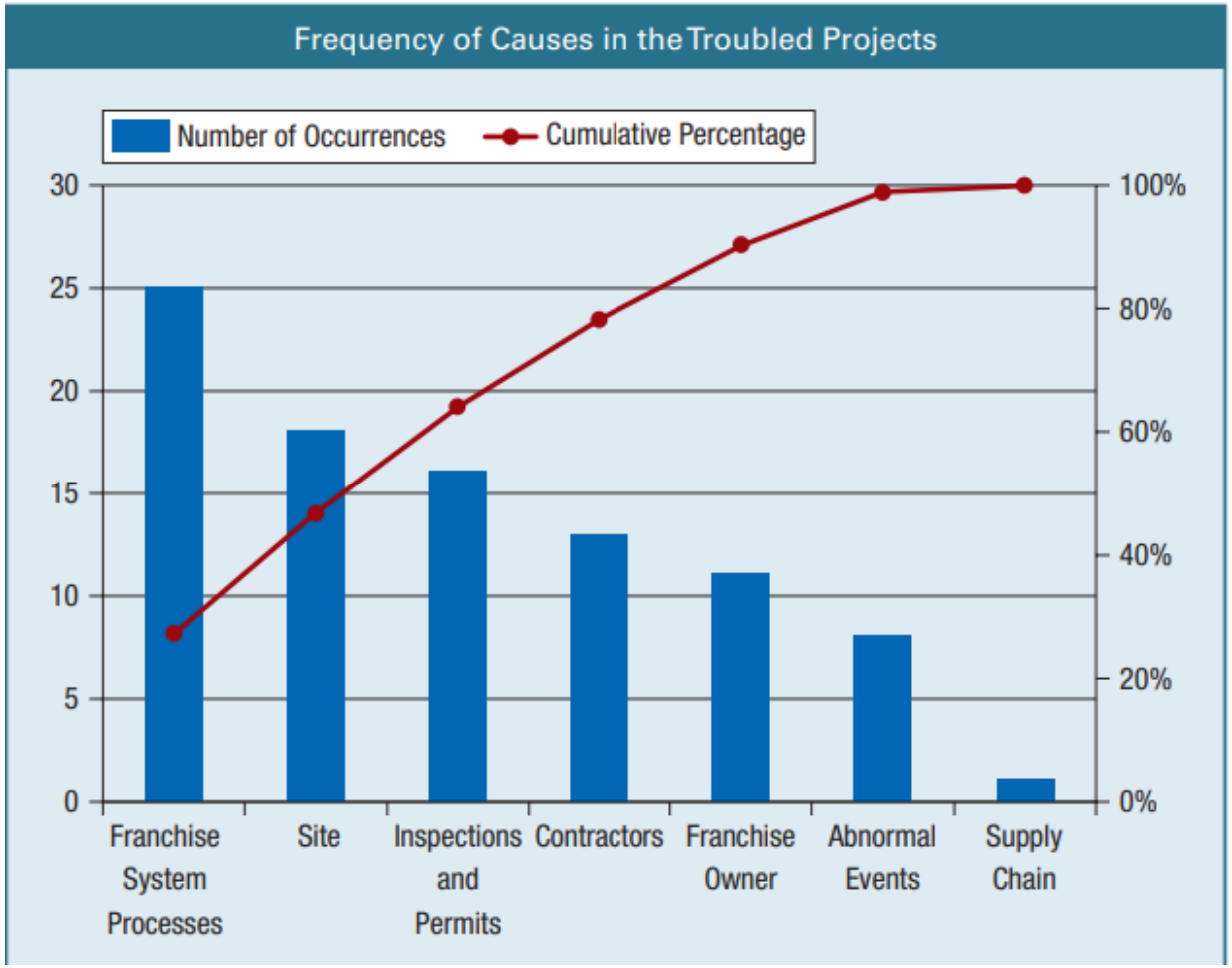
Exhibit 12.9

Days from Breaking Ground to Opening

Days	Frequency	Cumulative %
60	0	0%
80	1	1%
100	8	10%
120	26	41%
140	23	67%
160	11	80%
180	8	90%
200	5	95%
220	2	98%
240	1	99%
260	1	100%
280	0	100%
More	0	100%
Total	86	

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Exhibit 12.10



Source: Scott C Wright, PhD, P.E., PMP (University of Wisconsin-Platteville)

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12.2e Empowered Performance

- Have capable and willing workers at every level/function within a company.
- Corporate leaders develop the *organizational* culture.
- Project sponsors and managers develop the *project* culture.
- Recognize individuality:
  - *Projects benefit greatly from diversity in workforce*
  - *Project manager should discuss one-on-one individual's goals for project*
- Capitalize on individual strengths.
- Emphasize individual responsibilities.
- Use appropriate collaboration/share learning.

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Exhibit 12.11

Vintage Aircraft Shipping Project

Global Shipping Company (GSC) was approached by an individual who was interested in selling and shipping an antique $1 million 1942 Staggered-Wing Beech aircraft from Cincinnati to a buyer in Australia. Since the aircraft was fragile, a plan needed to be developed for moving it as economically as possible while avoiding damage.

One challenge was handling the entire project in-house using only the company's staff, equipment, and resources, and the other was devising a custom solution for moving this unusual piece of cargo.

GSC has an organizational culture that encourages cross-training, collaboration among departments, risk-taking, and designing creative approaches to problem solving while minimizing cost. Because of the size and fragility of the aircraft, a strategy was devised to dismantle it and ship via containerized ocean freight. The project was broken down into five distinct segments: pickup, dismantling, packing, loading, and shipping.

To pick up the entire aircraft, the equipment, permits, and escorts had to be arranged to get the aircraft intact from the airport and move it to the warehouse down a major street on the back of a flatbed truck. In order to fit in a standard ocean container, the aircraft had to be dismantled—under the supervision of the FAA—and documented to meet FAA regulations. To avoid damage, each piece had to be individually packaged. Different types of cloth and foam had to be tested and selected in order to prevent scratching the aircraft. Due to the height restrictions, the warehouse personnel had to design and build a custom gurney to allow the body of the plane to be wheeled into the container and secured. Once packaged, the individual pieces were then loaded, blocked, and braced into the container to prevent damage while in transit; then the aircraft was shipped.



Source: Danny McKee, Global Shipping Company.

The dismantling, documentation, and packing process was designed in a way that the new owner of the aircraft could replicate it in order to move the plane for air shows and events.

The project's success was achieved by having the courage to take on the project in the first place, the ability to use the company's resources creatively and efficiently, and the ability to adapt when unexpected events occurred. The result was a project that was successfully completed, meeting all FAA standards, exceeding stakeholder expectations, and developing a shipping process that can be replicated.

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Exhibit 12.12

Summary of Project Quality Core Concepts

Concept	Specific Guidance
Stakeholder Satisfaction	Identify all internal and external stakeholders. Prioritize among the stakeholders. Understand the prioritized stakeholders' requirements. Develop standards to ensure the requirements are met. Make trade-off decisions. Realize stakeholders will judge quality both of work processes and deliverables. Measure twice, cut once. (Plan and check the plan.) Meet requirements but exceed expectations. Develop capable customers.
Concept	**Specific Guidance**
Process Improvement	Learn about process with the supplier-input-process-output-customer model. Realize designing a quality process is far better than merely trying to find mistakes. Ensure project processes are capable and flexible. Control project processes to make them predictable. Improve project processes using a model based upon the plan-do-check-act concept.
Fact-Based Management	Understand the difference between common and special causes of variation. Select a few key well-defined items to measure. Carefully collect data and use appropriate analysis techniques to create useful information. Encourage truthful, transparent, and challenging communication when making decisions.
Empowered Performance	Develop capable and willing workers at every level and every function. Develop a risk-taking project culture. Understand each person is an individual. To the extent possible, let everyone do what they will enjoy doing and what their strengths support. Ensure everyone understands and accepts their responsibilities. Share lessons learned and other information as widely as possible.

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12.3 Plan Quality Management (1 of 2)

12.3a Quality Policy
12.3b Quality Management Plan Contents
12.3c Quality Baseline
12.3d Process Improvement Plan

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12.3 Plan Quality Management (2 of 2)

A **Quality Management Plan** is a component of the project management plan that describes how the project will perform in accordance with the quality policy.

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12.3a Quality Policy

Concise statements from top management to guide quality efforts may include:

- Reference to customers.
- Improving processes.
- Satisfying requirements.
- Products and services.
- Value to customers.
- Cost control.

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Exhibit 12.13

Evolution of Terms in Quality Policies though 2013, 2017, and 2021

Term	Percent of Policies in 2013	Percent of Policies in 2017	Percent of Policies in 2021
Customer	92	80	92
Improvement/Innovation	84	80	84
Employee Engagement	44	60	80
Satisfy Requirements/Meet Standards or Laws	68	72	60
Best/Excellent/High Quality/ Exceed Requirements	44	44	56
Suppliers/Collaboration	16	24	48

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12.3b Quality Management Plan Contents

- The project's overall quality objectives.
- Key project deliverables and the standards to evaluate each.
- Deliverable completeness and correctness criteria from customer's viewpoint.
- Quality control activities.
- Quality assurance activities.
- Quality tools.
- Quality roles and responsibilities.
- Quality reporting plan.

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12.3c Quality Baseline

- Reflects agreed-upon quality objectives.
- Include metrics:
  - *What will be measured.*
  - *How it will be measured.*
  - *The target value.*

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12.3d Process Improvement Plan

Details how to analyze work processes with an eye toward improving them.

A definition of “done” is explicitly stated.

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12.4 Manage Quality (1 of 2)

Manage Quality
Quality Assurance

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12.4 Manage Quality (2 of 2)

It is the process of using the quality plan and policy to perform tasks that will most likely lead to creating project outputs to customers' satisfaction.

A key component of **Manage Quality** is the forward-looking quality assurance.

Perform Quality Assurance is an executing process that deals with process improvements and ensures that deliverables are free of errors.

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Two Main Methods of Quality Assurance

1. **Quality Audit** is used to determine what methods are being used and whether they are effective.
2. **Process Analysis** is used to improve both quality and productivity.

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Quality Assurance (1 of 2)

Quality audits often lead to one or more of the following recommendations:

1. **Preventive actions**—taken to ensure future performance is acceptable.
2. **Corrective actions**—intended to bring performance back into alignment with baseline.
3. **Defect repair**—judgment needed to decide whether to repair or scrap and remake.

Process Analysis is breaking processes down into steps in order to determine ways to increase overall efficiency.

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Quality Assurance (2 of 2)

One common type of process analysis is benchmarking, which consists of the following:

1. Determine a process that needs dramatic improvement.
2. Identify another organization that performs that process very well.
3. Make a deal with that organization to learn from them *(they might require payment or the sharing of one of the observer's best practices with them).*
4. Determine what needs to be observed and what questions need to be asked.
5. Make a site visit to observe and question the other organization.
6. Decide which observed methods will help the organization.
7. Adapt the methods to fit the organization's culture and situation.
8. Try the new methods on a small scale.
9. Evaluate the results.
10. If the methods are good enough, adopt them.

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12.5 Control Quality (1 of 2)

Control Quality
Monitor the Project Quality
Quality Control Terms

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12.5 Control Quality (2 of 2)

Control Quality is the activity used to verify that deliverables are complete, correct, and acceptable; the purposes of **quality control** are to improve project processes and outputs, as well as to reduce the number of defects and inefficiencies.

- Test whether specific project deliverables meet their quality standards.
- Includes inspection of inputs, activities, deliverables, and a reporting system.

Agile creates and manages quality in small batch sizes; Agile projects may allow for more variability in quality processes—achieving a Minimum Viable Product (MVP)—in exchange for speed to market.

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Monitor Project Quality: General Lessons

- Conduct an **inspection** before a critical or expensive process to make sure the inputs are good before spending a large amount of money or time on them.
- Process **stages** in which one worker hands off work to another worker are good times for both the workers to conduct inspections.
- **Milestones** identified in the project charter provide good inspection points.
- As practiced in software development, think of an inspection in terms of **units** *(individual components)*, integration *(how components work together)*, and the system *(how the deliverable performs)*.

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Quality Control Terms

- Many terms with specific meanings are used in project quality control.
- While few projects repeat processes enough times to formally use statistical quality control, the concepts are still useful in making good decisions.

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Exhibit 12.14

Pairs of Project Quality Control Terms

Term:	Sometimes Confused with:
Prevention	Inspection
Sample	Population
Attribute	Variable
Precision	Accuracy
Tolerance	Control limit
Capable	In control
Special cause	Common cause
Preventive action	Corrective action

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12.6 Cost of Quality

Cost of Conformance
Cost of Nonconformance

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Exhibit 12.15

Costs of Quality

Cost of Conformance	Cost of Nonconformance
• Planning • Training • Process control and validation • Product design validation • Test and evaluation • Quality audits • Maintenance and calibration • Inspection • Field testing	• Scrap • Rework and repair • Additional material • Inventory • Warranty repairs and service • Compliant handling • Liability judgments • Product recalls • Field service • Expediting

Cost of Quality = Cost of conformance + Cost of nonconformance

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12.7 Develop Project Management Plan

12.7a Resolve Conflicts
12.7b Establish Configuration Management
12.7c Apply Sanity Tests to All Project Plans

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Questions to Resolve Conflicts

- What do they really want and need most from the project?
- Are all of the quality standards truly mandatory, or can one of them be relaxed a bit?
- Is the imposed deadline critical, or, considering the impact it poses for costs and risks, can it be relaxed a bit?
- Is the budget a true maximum, or can it be adjusted to secure the desired features?

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12.7b Establish Configuration Management

A **Configuration Management System** is composed of four parts:

1. Process for identifying and uniquely naming items to be controlled.
2. Activity of managing the project deliverables and documentation.
3. Recording and reporting all changes.
4. Verifying the correctness of all deliverables and their components.

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12.7c Apply Sanity Tests to All Project Plans



- *Does the critical path look reasonable?*
- *Do the milestones look achievable?*
- *Are some resources overallocated?*
- *Does everyone understand what they are supposed to do?*
- *Do we really understand our customers?*
- *Are the customers' desires likely to change?*
- *How well do we understand the standards we will be judged against?*
- *Are the methods for completing our work really sensible?*
- *Are we confident we can gather and analyze the data we need to control the plan?*

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12.8 Kickoff Project (1 of 2)

12.8a Preconditions to Meeting Success
12.8b Meeting Activities

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12.8 Kickoff Project (2 of 2)

- A formal kickoff gives stakeholders a chance to see how everything fits together.
- This step convinces project stakeholders that the project leaders will be good stewards of assets.
- All interested parties are eager to commit to the project and get on with the work.

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12.8a Preconditions to Meeting Success

- Sponsor and project manager set clear direction during the planning.
- Core team commits to the project first.
- Create an atmosphere of trust and relationship building.
- Project leaders practice active listening.
- Include as many as possible in planning to enhance buy-in.

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12.8b Meeting Activities

Typical activities included in the kickoff meeting are the following:

- The sponsor and project manager describing the importance of the project.
- The customer(s) describing their acceptance standards, sense of urgency, and budget concerns.
- The project manager outlining the project goals.
- The project manager and the core team describing work expectations.
- The project manager unfolding the project plan and its current status (if work has commenced).
- The core team explaining the communications, risk, and quality plans.
- Everyone asking questions and making suggestions.
- The project manager authorizing appropriate changes to the project plan.
- Everyone concurring with the overall plan and to their individual action items.

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Exhibit 12.16 (1 of 2)

IS&T Project Launch Assessment Agenda

Purpose: The Project Manager is to illustrate to an executive audience the chartered IS&T project's readiness to successfully launch. Upon conclusion of the Project Manager's presentation, the executive audience will determine and document the actions required for the project to launch.

Prerequisite: The Project Manager is required to complete the Project Deliverable Review and receive documented approval from the Project Deliverable Review Board in order to proceed to the Project Launch Assessment.

Standard Participants

- Core Group (CG) (CIO and IS&T Director) PMO Manager
- Project Manager
- Functional Manager
- Quality Consult
- Security Consult (Optional)
- Test Coordinator
- Sponsor

Required Documents: The Project Manager is required to present the PLA materials online. If a paper copy is needed, it should be printed double-sided.

- Project Charter PMO Risk Forms
- Project Financial Worksheet
- Master Test Plan
- Progress Report—PDR

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Exhibit 12.16 (2 of 2)

Project Launch Assessment Agenda: The Project Manager is required to present all of the listed deliverables in the provided order, focusing on specifically the identified components and content specified.

1. Project Charter—Discuss Business Need, Purpose, Logical Scope: In-Scope, Out-of-Scope, and Assumptions.
2. Master Test Plan—Discuss Sections 1.3—Test Levels, Objectives, and Deliverables; 3—Test Timeline and Key Events; and 5—Define System Characteristics, Relative Importance, and Subsystems.
3. Privacy and Security—Discuss the Security and HIPAA Template for PMO Projects.
4. Risk Forms—Discuss all populated and scored forms created to date.
5. Project Financial Worksheet—Discuss populated spreadsheet.
6. Progress Report (PDR)—Speak to the current status of all actions provided for each deliverable.

Source: Nancy D'Quila, PMP.

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12.9 Baseline and Communicate Project Management Plan (1 of 2)



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12.9 Baseline and Communicate Project Management Plan (2 of 2)

- A baseline is the approved plan.
- A project plan becomes official when enough info is available for key stakeholders to commit to and baseline the plan.
- Most of the planning is done, and most of the executing is just beginning.

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12.10 *PMBOK Guide 7e*

12.10a Stakeholders
12.10b Team
12.10c Work
12.10d Delivery
12.10e Measurement
12.10f Uncertainty

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12.10a Stakeholders

- To ensure stakeholder satisfaction, the project team should develop a collaborative and productive working relationship with stakeholders.
- Teams draw out stakeholder needs, desires, requirements, and acceptance criteria.

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12.10b Team

Effective project leaders establish and maintain the project vision, use critical thinking, and strive to understand and improve the motivation of their team members.

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12.10c Work

Project managers need to balance competing constraints and the workload of their teams to consistently create deliverables that will deliver the desired quality and value.

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12.10d Delivery

- Delivery quality focuses on value as determined by required performance levels.
- Quality costs include prevention, appraisal, internal failure, and external failure.

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12.10e Measurement

- Metrics directly related to quality include defects, performance measures, work in progress, and stakeholder satisfaction.
- Each metric should include the desired baseline and the threshold beyond which corrective action is taken.

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12.10f Uncertainty

To respond to uncertainty—gather information, prepare for multiple outcomes, investigate alternative designs, and build-in resilience.

- *Deal with conceptual ambiguity by using clear definitions, and for situational ambiguity use progressive elaboration, experiments, and prototypes.*
- *Deal with complexity by decoupling, simulating, viewing from diverse perspectives, balancing data used, iterating, engaging, and building in redundancy.*
- *Deal with volatility by analyzing alternatives and using reserves.*

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12.11 Agile Projects

12.11a Agile Terms

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Exhibit 12.17

Comparison of Plan-Driven and Agile Approaches for Project Quality Planning and Project KickOff

Project Quality Planning and Project KickOff Questions	Plan-Driven	Agile
What framework is used for broad quality guidance?	Malcom Baldrige and/or ISO	Malcom Baldrige and/or ISO
What core project quality concepts are used as guidance?	Stakeholder satisfaction, process management, fact-based management, and empowered performance	Stakeholder satisfaction, process management, fact-based management, and empowered performance
What organizational level guidance is used?	Parent organization's quality policy and industry quality standards	Parent organization's quality policy and industry quality standards
When and how frequently is quality inspected?	Varies	Early and often
When and how are requirements changed?	Seldom, through change control and contracts	Frequently, through refactoring and continuous integration
When and how is the project kicked off?	After significant planning, with kickoff meeting baselining project	Early, after chartering, with team identifying personas, creating a backlog, and planning the first release

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12.11a Agile Terms (1 of 2)

TERM	DEFINITION
Build quality in	*Workflow is reduced and waste increased if mistakes are made.*
Continuous improvement	*Pursue perfection by enlisting all in never-ending effort to get better.*
Make process explicit	*Ensure everyone understands how a process works.*
Value stream mapping	*Visual flowchart identifying value and nonvalue activities.*
Inspection	*Everyone examines, trying to improve product, process, and people.*
Testing	*Evaluating to ensure that the deliverable meets requirements, which serves as an important feedback mechanism.*
Test-driven development	*The practice of writing tests to be used before writing code.*

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12.11a Agile Terms (2 of 2)

TERM	DEFINITION
Test plan	*Determine how you will verify work is acceptable before performing.*
Unit test	*Ensuring small function or classes work properly.*
Integration test	*Ensuring new functionality works immediately in existing system.*
Acceptance test	*Verifying for customer entire system works correctly.*
Continuous integration	*The practice of immediately ensuring each bit works in the system.*
Refactoring	*Simplifying and cleaning a product to make it stable and maintainable.*

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12.12 Using MS Project for Project Baselines

12.12a Baseline the Project Plan
12.12b Create the First Time Baseline
12.12c Subsequent Baselines
12.12d Viewing Baselines and Variances

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12.12a Baseline the Project Plan

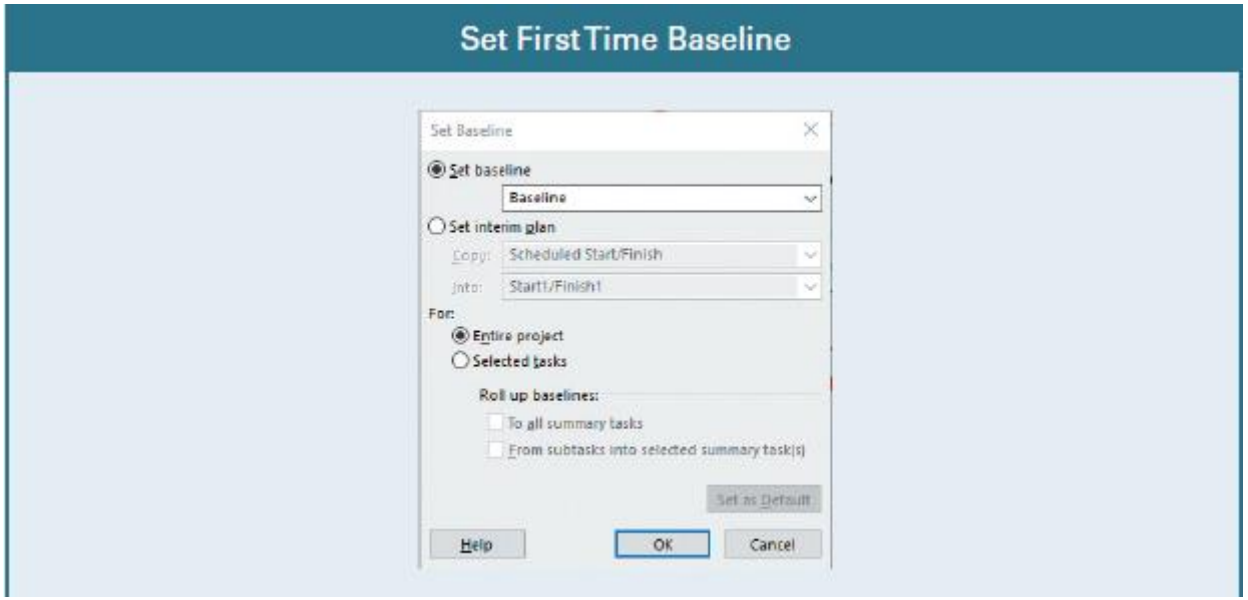
Once project execution begins, the *actual* value of the inputs will begin to vary from the *planned* value.

MS Project will track the variance between baseline and actual values to help the project manager understand project performance.

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Exhibit 12.18

Project Tab>> Schedule Groups>>
click **Set Baseline** >>
click **OK.**



Source: Microsoft Corporation.

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12.12c Subsequent Baselines (1 of 2)

If a change is approved, the tasks must be re-baselined, using the following steps:

1. Select the changed or added activities, milestones, and WBS elements.
2. Click **Project tab**>>**Schedule Group**>> click **Set Baseline**, then click **Set Baseline …**
3. Ensure the original baseline is selected in drop-down menu under **"Set Baseline."**
4. Click **Selected tasks**>>check **To all summary tasks.**
5. Click **OK**.

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12.12c Subsequent Baselines (2 of 2)

Reasons to change the baseline:

- Changes to the project scope.
- Project delay or restart.
- Unavailability of planned resources.
- Slower cash flow than planned.
- Occurrence of risk events.
- Quality problems.

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12.12d Viewing Baselines and Variances

To display variances in Gantt Chart View:

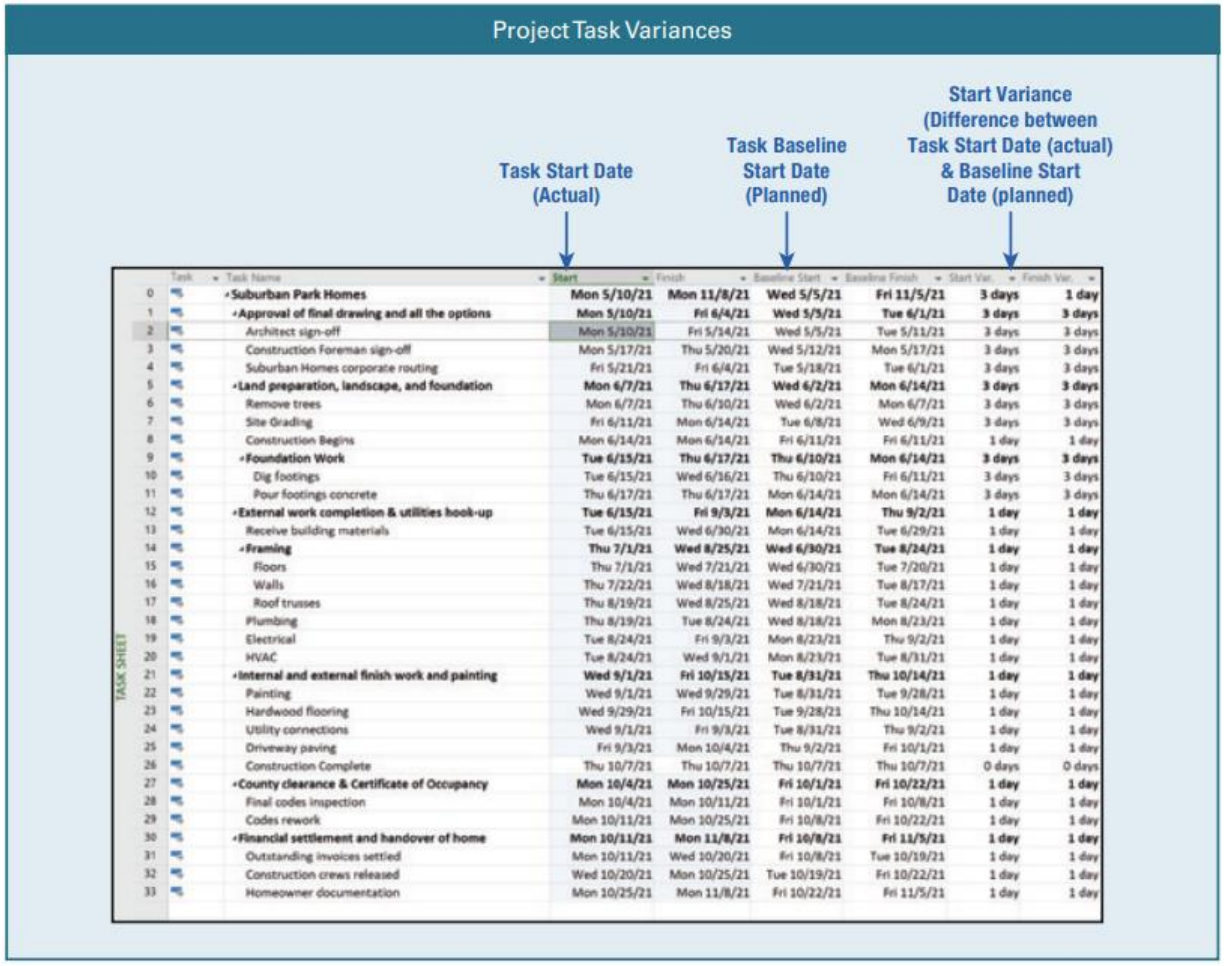
- Click the **View Tab**>>**Task Views**>>**Gantt Chart.**
- Click the **Format Tab**>>**Bar Styles Group**>>**Baseline>>**click the **Baseline you want to view.**

To view actual and baseline values side by side:

- Click the **View Tab**>>**Task Views**>>click **Other Views**>>**Task Sheet.**
- In the **Data Group**>>**Tables**>>**Variances.**

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Exhibit 12.19



	Task	Task Name	Start	Finish	Baseline Start	Baseline Finish	Start Var.	Finish Var.
0		**Suburban Park Homes**	**Mon 5/10/21**	**Mon 11/8/21**	**Wed 5/5/21**	**Fri 11/5/21**	**3 days**	**1 day**
1		**Approval of final drawing and all the options**	**Mon 5/10/21**	**Fri 6/4/21**	**Wed 5/5/21**	**Tue 6/1/21**	**3 days**	**3 days**
2		Architect sign-off	Mon 5/10/21	Fri 5/14/21	Wed 5/5/21	Tue 5/11/21	3 days	3 days
3		Construction Foreman sign-off	Mon 5/17/21	Thu 5/20/21	Wed 5/12/21	Mon 5/17/21	3 days	3 days
4		Suburban Homes corporate routing	Fri 5/21/21	Fri 6/4/21	Tue 5/18/21	Tue 6/1/21	3 days	3 days
5		**Land preparation, landscape, and foundation**	**Mon 6/7/21**	**Thu 6/17/21**	**Wed 6/2/21**	**Mon 6/14/21**	**3 days**	**3 days**
6		Remove trees	Mon 6/7/21	Thu 6/10/21	Wed 6/2/21	Mon 6/7/21	3 days	3 days
7		Site Grading	Fri 6/11/21	Mon 6/14/21	Tue 6/8/21	Wed 6/9/21	3 days	3 days
8		Construction Begins	Mon 6/14/21	Mon 6/14/21	Fri 6/11/21	Fri 6/11/21	1 day	1 day
9		**Foundation Work**	**Tue 6/15/21**	**Thu 6/17/21**	**Thu 6/10/21**	**Mon 6/14/21**	**3 days**	**3 days**
10		Dig footings	Tue 6/15/21	Wed 6/16/21	Thu 6/10/21	Fri 6/11/21	3 days	3 days
11		Pour footings concrete	Thu 6/17/21	Thu 6/17/21	Mon 6/14/21	Mon 6/14/21	3 days	3 days
12		**External work completion & utilities hook-up**	**Tue 6/15/21**	**Fri 9/3/21**	**Mon 6/14/21**	**Thu 9/2/21**	**1 day**	**1 day**
13		Receive building materials	Tue 6/15/21	Wed 6/30/21	Mon 6/14/21	Tue 6/29/21	1 day	1 day
14		**Framing**	**Thu 7/1/21**	**Wed 8/25/21**	**Wed 6/30/21**	**Tue 8/24/21**	**1 day**	**1 day**
15		Floors	Thu 7/1/21	Wed 7/21/21	Wed 6/30/21	Tue 7/20/21	1 day	1 day
16		Walls	Thu 7/22/21	Wed 8/18/21	Wed 7/21/21	Tue 8/17/21	1 day	1 day
17		Roof trusses	Thu 8/19/21	Wed 8/25/21	Wed 8/18/21	Tue 8/24/21	1 day	1 day
18		Plumbing	Thu 8/19/21	Tue 8/24/21	Wed 8/18/21	Mon 8/23/21	1 day	1 day
19		Electrical	Tue 8/24/21	Fri 9/3/21	Mon 8/23/21	Thu 9/2/21	1 day	1 day
20		HVAC	Tue 8/24/21	Wed 9/1/21	Mon 8/23/21	Tue 8/31/21	1 day	1 day
21		**Internal and external finish work and painting**	**Wed 9/1/21**	**Fri 10/15/21**	**Tue 8/31/21**	**Thu 10/14/21**	**1 day**	**1 day**
22		Painting	Wed 9/1/21	Wed 9/29/21	Tue 8/31/21	Tue 9/28/21	1 day	1 day
23		Hardwood flooring	Wed 9/29/21	Fri 10/15/21	Tue 9/28/21	Thu 10/14/21	1 day	1 day
24		Utility connections	Wed 9/1/21	Fri 9/3/21	Tue 8/31/21	Thu 9/2/21	1 day	1 day
25		Driveway paving	Fri 9/3/21	Mon 10/4/21	Thu 9/2/21	Fri 10/1/21	1 day	1 day
26		Construction Complete	Thu 10/7/21	Thu 10/7/21	Thu 10/7/21	Thu 10/7/21	0 days	0 days
27		**County clearance & Certificate of Occupancy**	**Mon 10/4/21**	**Mon 10/25/21**	**Fri 10/1/21**	**Fri 10/22/21**	**1 day**	**1 day**
28		Final codes inspection	Mon 10/4/21	Mon 10/11/21	Fri 10/1/21	Fri 10/8/21	1 day	1 day
29		Codes rework	Mon 10/11/21	Mon 10/25/21	Fri 10/8/21	Fri 10/22/21	1 day	1 day
30		**Financial settlement and handover of home**	**Mon 10/11/21**	**Mon 11/8/21**	**Fri 10/8/21**	**Fri 11/5/21**	**1 day**	**1 day**
31		Outstanding invoices settled	Mon 10/11/21	Wed 10/20/21	Fri 10/8/21	Tue 10/19/21	1 day	1 day
32		Construction crews released	Wed 10/20/21	Mon 10/25/21	Tue 10/19/21	Fri 10/22/21	1 day	1 day
33		Homeowner documentation	Mon 10/25/21	Mon 11/8/21	Fri 10/22/21	Fri 11/5/21	1 day	1 day

TASK SHEET

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Summary (1 of 4)

- The contemporary approach to project quality draws on contributions of Deming, Juran, and others.
- The Malcolm Baldrige award, ISO certification, and Six Sigma provide quality frameworks.
- The quality plan should include the quality baseline, which defines performance expectations.
- Once the quality plan is complete, hold a kickoff meeting with all project stakeholders.
- The kickoff meeting signifies the official end of planning and beginning of the project execution stage.

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Summary (2 of 4)

First concept in contemporary project quality → *stakeholder satisfaction*.

- Understand project stakeholders
- Prioritize stakeholder needs
- Manage toward stakeholder needs
- Keep the relationships strong
- Strive to ensure the customer is capable of using project deliverables.

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Summary (3 of 4)

Second concept → *process management.*

- Understand forms of improvement.
- Seek the root cause of problems.
- Use an appropriate model to guide improvement efforts (DMAIC).

Third concept → *fact-based management.*

- Understand variation.
- Make good decisions regarding what to measure.
- Capture and analyze data correctly.
- Use information in an open, honest decision-making manner.

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Summary (4 of 4)

Fourth concept → *empowered performance*

- Capable and willing workers.
- Treat each person as an individual.
- Ensure people accept responsibility.
- Strive for collaboration.

In Quality Management Planning, either adopt the quality policy of the parent organization or supplement it.

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